

EEE Final Exam Important Math's:

Sure 15+ Mark

Chapter 01-FET – 19.1-19.3/19.5

Field Effect Transistors ■ 515

Example 19.1. Fig. 19.14 shows the transfer characteristic curve of a JFET. Write the equation for drain current.

Solution. Referring to the transfer characteristic curve in Fig. 19.14, we have,

$$\begin{aligned} I_{DSS} &= 12 \text{ mA} \\ V_{GS(off)} &= -5 \text{ V} \\ \therefore I_D &= I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2 \\ \text{or } I_D &= 12 \left[1 + \frac{V_{GS}}{5} \right]^2 \text{ mA Ans.} \end{aligned}$$

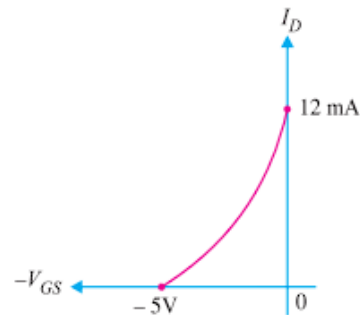


Fig. 19.14

Example 19.2. A JFET has the following parameters: $I_{DSS} = 32 \text{ mA}$; $V_{GS(off)} = -8 \text{ V}$; $V_{GS} = -4.5 \text{ V}$. Find the value of drain current.

$$\begin{aligned} \text{Solution. } I_D &= I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2 \\ &= 32 \left[1 - \frac{(-4.5)}{-8} \right]^2 \text{ mA} \\ &= 6.12 \text{ mA} \end{aligned}$$

Example 19.3. A JFET has a drain current of 5 mA. If $I_{DSS} = 10 \text{ mA}$ and $V_{GS(off)} = -6 \text{ V}$, find the value of (i) V_{GS} and (ii) V_P .

$$\begin{aligned} \text{Solution. } I_D &= I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2 \\ \text{or } 5 &= 10 \left[1 + \frac{V_{GS}}{6} \right]^2 \\ \text{or } 1 + \frac{V_{GS}}{6} &= \sqrt{5/10} = 0.707 \\ \text{(i) } \therefore V_{GS} &= -1.76 \text{ V} \\ \text{(ii) and } V_P &= -V_{GS(off)} = 6 \text{ V} \end{aligned}$$

Example 19.4. For the JFET in Fig. 19.15, $V_{GS} = -4 \text{ V}$ and $I_D = 12 \text{ mA}$. Determine the

Fig. 19.15

Fig. 19.16

Example 19.5. Determine the value of drain current for the circuit shown in Fig. 19.16.

Solution. It is clear from Fig. 19.16 that $V_{GS} = -2 \text{ V}$. The drain current for the circuit is given by;

$$\begin{aligned} I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2 \\ &= 3 \text{ mA} \left(1 - \frac{-2 \text{ V}}{-6 \text{ V}} \right)^2 \\ &= (3 \text{ mA}) (0.444) = 1.33 \text{ mA} \end{aligned}$$

Example 19.4. For the JFET in Fig. 19.15, $V_{GS(off)} = -4V$ and $I_{DSS} = 12\text{ mA}$. Determine the minimum value of V_{DD} required to put the device in the constant-current region of operation.

Solution. Since $V_{GS(off)} = -4V$, $V_P = 4V$. The minimum value of V_{DS} for the JFET to be in constant-current region is

$$V_{DS} = V_P = 4V$$

In the constant current region with $V_{GS} = 0V$,

$$I_D = I_{DSS} = 12\text{ mA}$$

Applying Kirchhoff's voltage law around the drain circuit, we have,

$$\begin{aligned} V_{DD} &= V_{DS} + V_{R_D} = V_{DS} + I_D R_D \\ &= 4V + (12\text{ mA})(560\Omega) = 4V + 6.72V = \mathbf{10.72V} \end{aligned}$$

This is the value of V_{DD} to make $V_{DS} = V_P$ and put the device in the constant-current region.

Chapter - Switching transistors: -18.2, 18.3 (extra 18.1,18.4)



Electronic Switches

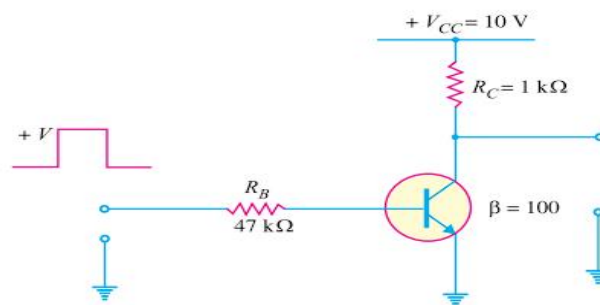
Example 18.1. Determine the minimum high input voltage ($+V$) required to saturate the transistor switch shown in Fig. 18.6.

Solution. Assuming the transistor to be ideal,

$$I_{C(sat)} = V_{CC}/R_C = 10\text{ V}/1\text{ k}\Omega = 10\text{ mA}$$

$$\therefore I_B = \frac{I_{C(sat)}}{\beta} = \frac{10\text{ mA}}{100} = 0.1\text{ mA}$$

$$\begin{aligned} \text{Now } +V &= I_B R_B + V_{BE} \\ &= (0.1\text{ mA})(47\text{ k}\Omega) + 0.7 \\ &= 4.7 + 0.7 = \mathbf{5.4\text{ V}} \end{aligned}$$



Example 18.2. A transistor is used as a switch. If $V_{CC} = 10V$, $R_C = 1\text{ k}\Omega$ and $I_{CBO} = 10\text{ }\mu\text{A}$, determine the value of V_{CE} when the transistor is (i) cut off and (ii) saturated.

Solution.

(i) At cut off

$$\begin{aligned} I_C &= I_{CBO} = 10\text{ }\mu\text{A} \\ \therefore V_{CE} &= V_{CC} - I_{CBO} R_C \\ &= 10\text{ V} - 10\text{ }\mu\text{A} \times 1\text{ k}\Omega = 10\text{V} - 10\text{ mV} = \mathbf{9.99V} \end{aligned}$$

(ii) At saturation

$$\begin{aligned} I_{C(sat)} &= \frac{V_{CC} - V_{knee}}{R_C} \\ \therefore V_{CE} &= V_{knee} = \mathbf{0.7V} \end{aligned}$$

Example 18.3. Fig. 18.9 shows the transistor switching circuit. Given that $R_B = 2.7\text{ k}\Omega$, $V_{BB} = 2V$, $V_{BE} = 0.7V$ and $V_{knee} = 0.7V$.

(i) Calculate the minimum value of β for saturation.

(ii) If V_{BB} is changed to $1V$ and transistor has minimum $\beta = 50$, will the transistor be saturated.

Solution.

$$\begin{aligned} \text{(i)} \quad I_B &= \frac{V_{BB} - V_{BE}}{R_B} \\ &= \frac{2V - 0.7V}{2.7\text{ k}\Omega} = \frac{1.3\text{ V}}{2.7\text{ k}\Omega} = 0.48\text{ mA} \\ \text{Now} \quad I_{C(sat)} &= \frac{V_{CC} - V_{knee}}{R_C} \end{aligned}$$

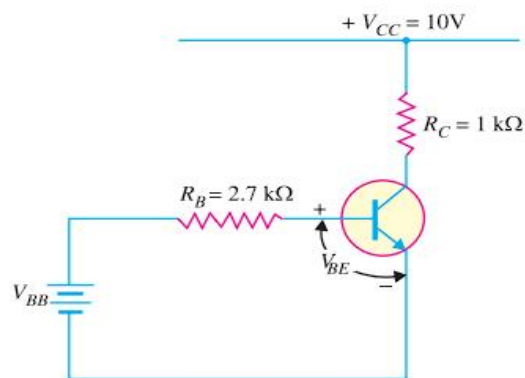


Fig. 18.9

$$\begin{aligned} &= \frac{10V - 0.7V}{1\text{ k}\Omega} = \frac{9.3V}{1\text{ k}\Omega} = 9.3\text{ mA} \\ \therefore \text{Minimum } \beta &= \frac{I_{C(sat)}}{I_B} = \frac{9.3\text{ mA}}{0.48\text{ mA}} = \mathbf{19.4} \\ \text{(ii)} \quad I_B &= \frac{V_{BB} - V_{BE}}{R_B} \\ &= \frac{1V - 0.7V}{2.7\text{ k}\Omega} = \frac{0.3V}{2.7\text{ k}\Omega} = 0.111\text{ mA} \\ \therefore I_C &= \beta I_B = 50 \times 0.111 = 5.55\text{ mA} \end{aligned}$$

Since the collector current is less than saturation current ($= 9.3\text{ mA}$), **the transistor will not be saturated.**

Example 18.4. In the astable multivibrator shown in Fig. 18.13, $R_2 = R_3 = 10 \text{ k}\Omega$ and $C_1 = C_2 = 0.01 \text{ }\mu\text{F}$. Determine the time period and frequency of the square wave.

Solution.

Here $R = 10 \text{ k}\Omega = 10^4 \text{ }\Omega$; $C = 0.01 \text{ }\mu\text{F} = 10^{-8} \text{ F}$

Time period of the square wave is

$$\begin{aligned} T &= 1.4 RC = 1.4 \times 10^4 \times 10^{-8} \text{ second} \\ &= 1.4 \times 10^{-4} \text{ second} = 1.4 \times 10^{-4} \times 10^3 \text{ m sec} \\ &= \mathbf{0.14 \text{ m sec}} \end{aligned}$$

Frequency of the square wave is

$$\begin{aligned} f &= \frac{1}{T \text{ in second}} \text{ Hz} = \frac{1}{1.4 \times 10^{-4}} \text{ Hz} \\ &= 7 \times 10^3 \text{ Hz} = \mathbf{7 \text{ kHz}} \end{aligned}$$

Chapter-Op-amp: 25.32-25.34, 25.44-25.47

Example 25.32. Calculate the output voltage from the noninverting amplifier circuit shown in Fig. 25.57 for an input of $120 \text{ }\mu\text{V}$.

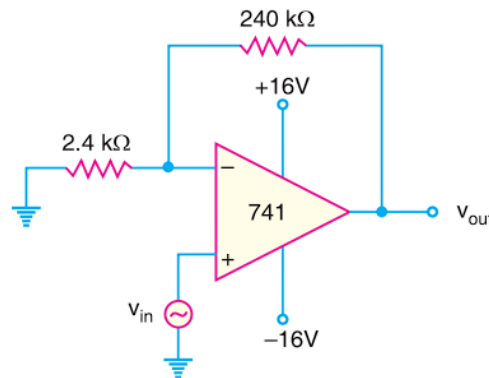


Fig. 25.57

Solution. Voltage gain, $A_{CL} = 1 + \frac{R_f}{R_i} = 1 + \frac{240 \text{ k}\Omega}{2.4 \text{ k}\Omega} = 1 + 100 = 101$

Output voltage, $v_{out} = A_{CL} \times v_{in} = (101) \times (120 \text{ }\mu\text{V}) = \mathbf{12.12 \text{ mV}}$

Example 25.33. For the noninverting amplifier circuit shown in Fig. 25.58, find the output voltage for an input voltage of (i) 1 V (ii) -1 V .

Solution. Voltage gain, $A_{CL} = 1 + \frac{R_f}{R_i} = 1 + \frac{10 \text{ k}\Omega}{1 \text{ k}\Omega} = 1 + 10 = 11$

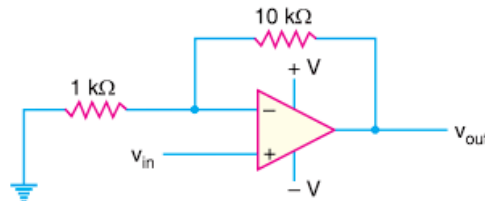


Fig. 25.58

(i) For $v_{in} = 1 \text{ V}$; $v_{out} = A_{CL} \times v_{in} = 11 \times 1 \text{ V} = \mathbf{11 \text{ V}}$

(ii) For $v_{in} = -1 \text{ V}$; $v_{out} = A_{CL} \times v_{in} = 11 \times (-1 \text{ V}) = \mathbf{-11 \text{ V}}$

Example 25.34. For the noninverting amplifier circuit shown in Fig. 25.59, find peak-to-peak output voltage.

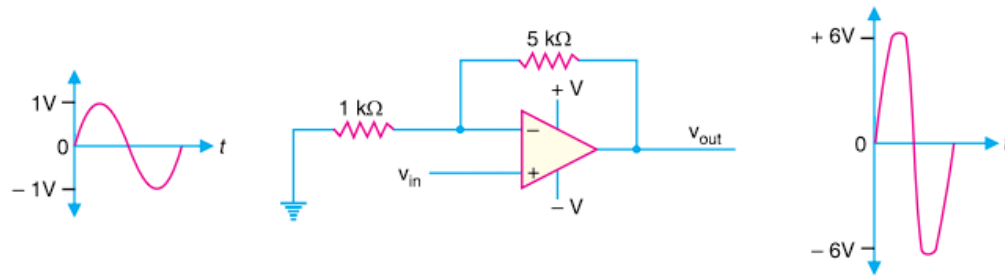


Fig. 25.59

Solution. The input signal is 2 V peak-to-peak.

$$\text{Voltage gain, } A_{CL} = 1 + \frac{R_f}{R_i} = 1 + \frac{5 \text{ k}\Omega}{1 \text{ k}\Omega} = 1 + 5 = 6$$

$$\therefore \text{Peak-to-peak output voltage} = A_{CL} \times v_{inpp} = 6 \times 2 = \mathbf{12 \text{ V}}$$

Example 25.44. Determine the output voltage for the summing amplifier in Fig. 25.72.

Solution. Referring to Fig. 25.72, all the three input resistor values are equal and each is equal to the value of feedback resistor. Therefore, the gain of the summing amplifier is 1. As a result, the output voltage is the algebraic sum of three input voltages.

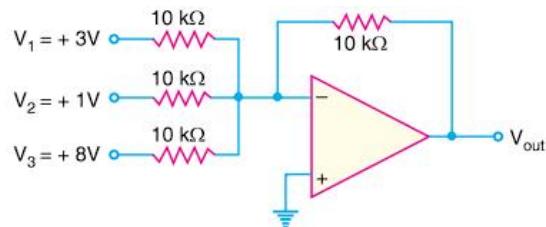


Fig. 25.72

$$\therefore V_{out} = -(V_1 + V_2 + V_3) = -(3 + 1 + 8) = \mathbf{-12 \text{ V}}$$

Example 25.45. Determine the output voltage for the summing amplifier shown in Fig. 25.73.

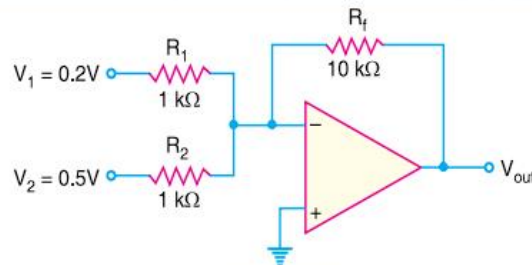


Fig. 25.73

Solution. $R_f = 10 \text{ k}\Omega$ and $R_1 = R_2 = R = 1 \text{ k}\Omega$. Therefore, gain of the amplifier $= -R_f/R = -10 \text{ k}\Omega/1 \text{ k}\Omega = -10$.

$$\text{Now } V_{out} = -\frac{R_f}{R}(V_1 + V_2) = -\frac{10 \text{ k}\Omega}{1 \text{ k}\Omega}(0.2 + 0.5) = \mathbf{-7 \text{ V}}$$

Note that the output voltage is not equal to the sum of input voltages. Rather it is equal to the sum of input voltages multiplied by the amplifier gain. In other words, the output voltage is proportional to the sum of the input voltages.

proportional to the sum of the input voltages.

Example 25.46. Determine the output voltage for the summing amplifier shown in Fig. 25.74.

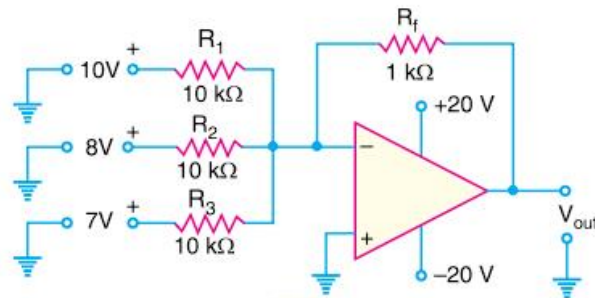


Fig. 25.74

Solution. $R_f = 1 \text{ k}\Omega$ and $R_1 = R_2 = R_3 = R = 10 \text{ k}\Omega$. Therefore, gain of the amplifier $= -R_f/R = -1 \text{ k}\Omega/10 \text{ k}\Omega = -1/10$.

$$\text{Now} \quad V_{out} = -\frac{R_f}{R} (V_1 + V_2 + V_3) = -\frac{1 \text{ k}\Omega}{10 \text{ k}\Omega} (10 + 8 + 7) = -2.5 \text{ V}$$

Note that output voltage of the amplifier is not equal to the sum of input voltages, but rather is proportional to the sum of input voltages. In this case, it is equal to one-tenth of the input sum. Picking random values of V_1 , V_2 and V_3 will show that this circuit always provides an output voltage that is one-tenth of the sum of input voltages.

Example 25.47. Two voltages of $+0.6 \text{ V}$ and -1.4 V are applied to the two input resistors of a summing amplifier. The respective input resistors are $400 \text{ k}\Omega$ and $100 \text{ k}\Omega$ and feedback resistor is $200 \text{ k}\Omega$. Determine the output voltage.

Solution. The output voltage of the summing amplifier is given by:

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

Here $R_f = 200 \text{ k}\Omega$; $R_1 = 400 \text{ k}\Omega$; $R_2 = 100 \text{ k}\Omega$; $V_1 = +0.6 \text{ V}$; $V_2 = -1.4 \text{ V}$

$$\therefore V_{out} = -200 \text{ k}\Omega \left(\frac{0.6}{400 \text{ k}\Omega} + \frac{-1.4}{100 \text{ k}\Omega} \right) = 2.5 \text{ V}$$

Note that a summing amplifier produces an output voltage that is proportional to the algebraic sum of the input voltages.

Chapter-Feedback: 13.2-13.5

Example 13.4. With a negative voltage feedback, an amplifier gives an output of 10 V with an input of 0.5 V . When feedback is removed, it requires 0.25 V input for the same output. Calculate (i) gain without feedback (ii) feedback fraction m_v

Solution.

(i) Gain without feedback, $A_v = 10/0.25 = 40$

(ii) Gain with feedback, $A_{vf} = 10/0.5 = 20$

$$\begin{aligned}
 \text{Now} \quad A_{vf} &= \frac{A_v}{1 + A_v m_v} \\
 \text{or} \quad 20 &= \frac{40}{1 + 40 m_v} \\
 \text{or} \quad 20 + 800 m_v &= 40 \\
 \text{or} \quad m_v &= \frac{40 - 20}{800} = \frac{1}{40}
 \end{aligned}$$

Example 13.2. The overall gain of a multistage amplifier is 140. When negative voltage feedback is applied, the gain is reduced to 17.5. Find the fraction of the output that is feedback to the input.

Solution. $A_v = 140$, $A_{vf} = 17.5$

Let m_v be the feedback fraction. Voltage gain with negative feedback is

$$\begin{aligned}
 A_{vf} &= \frac{A_v}{1 + A_v m_v} \\
 \text{or} \quad 17.5 &= \frac{140}{1 + 140 m_v} \\
 \text{or} \quad 17.5 + 2450 m_v &= 140 \\
 \therefore m_v &= \frac{140 - 17.5}{2450} = \frac{1}{20}
 \end{aligned}$$

Example 13.3. When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50.

(i) Calculate the fraction of the output voltage feedback.

(ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

Solution.

(i) Gain without feedback, $A_v = 100$

Gain with feedback, $A_{vf} = 50$

Let m_v be the fraction of the output voltage feedback.

$$\begin{aligned}
 \text{Now} \quad A_{vf} &= \frac{A_v}{1 + A_v m_v} \\
 \text{or} \quad 50 &= \frac{100}{1 + 100 m_v} \\
 \text{or} \quad 50 + 5000 m_v &= 100 \\
 \text{or} \quad m_v &= \frac{100 - 50}{5000} = 0.01
 \end{aligned}$$

(ii) $A_{vf} = 75$; $m_v = 0.01$; $A_v = ?$

$$\begin{aligned}
 A_{vf} &= \frac{A_v}{1 + A_v m_v} \\
 \text{or} \quad 75 &= \frac{A_v}{1 + 0.01 A_v} \\
 \text{or} \quad 75 + 0.75 A_v &= A_v \\
 \therefore A_v &= \frac{75}{1 - 0.75} = 300
 \end{aligned}$$

Example 13.5. The gain of an amplifier without feedback is 50 whereas with negative voltage feedback, it falls to 25. If due to ageing, the amplifier gain falls to 40, find the percentage reduction in stage gain (i) without feedback and (ii) with negative feedback.

Solution.
$$A_{vf} = \frac{A_v}{1 + A_v m_v}$$

or
$$25 = \frac{50}{1 + 50 m_v}$$

or
$$m_v = 1/50$$

(i) **Without feedback.** The gain of the amplifier without feedback is 50. However, due to ageing, it falls to 40.

$$\therefore \text{ \%age reduction in stage gain} = \frac{50 - 40}{50} \times 100 = 20\%$$

(ii) **With negative feedback.** When the gain without feedback was 50, the gain with negative feedback was 25. Now the gain without feedback falls to 40.

$$\therefore \text{ New gain with negative feedback} = \frac{A_v}{1 + A_v m_v} = \frac{40}{1 + (40 \times 1/50)} = 22.2$$

$$\therefore \text{ \%age reduction in stage gain} = \frac{25 - 22.2}{25} \times 100 = 11.2\%$$

Example 14.5. Calculate the (i) operating frequency and (ii) feedback fraction for Hartley oscillator shown in Fig. 14.15. The mutual inductance between the coils, $M = 20 \mu\text{H}$.

Solution.

(i)
$$L_1 = 1000 \mu\text{H}; \quad L_2 = 100 \mu\text{H}; \quad M = 20 \mu\text{H}$$

$$\therefore \text{ Total inductance, } L_T = L_1 + L_2 + 2M$$

$$= 1000 + 100 + 2 \times 20 = 1140 \mu\text{H} = 1140 \times 10^{-6} \text{H}$$

$$\text{Capacitance, } C = 20 \text{ pF} = 20 \times 10^{-12} \text{ F}$$

* Referring to Fig. 14.14, it is clear that L_1 and L_2 are in series. Therefore, total inductance L_T is given by : $L_T = L_1 + L_2 + 2M$

** Referring to Fig. 14.14, the circulating current for the two inductors is the same. Further, inductive reactance is directly proportional to inductance.

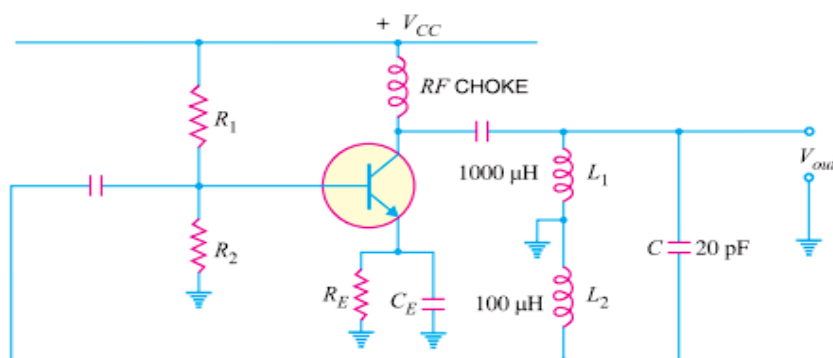


Fig. 14.15

$$\therefore \text{ Operating frequency, } f = \frac{1}{2\pi \sqrt{L_T C}} = \frac{1}{2\pi \sqrt{1140 \times 10^{-6} \times 20 \times 10^{-12}}} \text{ Hz}$$

$$= 1052 \times 10^3 \text{ Hz} = 1052 \text{ kHz}$$

(ii)
$$\text{ Feedback fraction, } m_v = \frac{L_2}{L_1} = \frac{100 \mu\text{H}}{1000 \mu\text{H}} = 0.1$$

$$L_1 = 1000 \mu\text{H}$$

Example 14.6. A 1 pF capacitor is available. Choose the inductor values in a Hartley oscillator so that $f = 1 \text{ MHz}$ and $m_v = 0.2$.

Solution.

$$\text{Feedback fraction, } m_v = \frac{L_2}{L_1}$$

$$\text{or } 0.2 = \frac{L_2}{L_1} \quad \therefore L_1 = 5L_2$$

$$\text{Now } f = \frac{1}{2\pi \sqrt{L_T C}}$$

$$\text{or } L_T = \frac{1}{C (2\pi f)^2} = \frac{1}{(1 \times 10^{-12}) (2\pi \times 1 \times 10^6)^2}$$

$$= 25.3 \times 10^{-3} \text{ H} = 25.3 \text{ mH}$$

$$\text{or } L_1 + L_2 = 25.3 \text{ mH} \quad (\because L_T = L_1 + L_2)$$

$$\text{or } 5L_2 + L_2 = 25.3 \quad \therefore L_2 = 25.3/6 = 4.22 \text{ mH}$$

$$\text{and } L_1 = 5L_2 = 5 \times 4.22 = 21.1 \text{ mH}$$

Prepared By

Mohammad Shakwath (C233257)